Movement graph - MRU

Example Problem with Uniform Motion (Constant Velocity) Graphs

Problem Statement

A girl walks from her house to school, which is 1 km away, and it takes her 20 minutes. She stays there for 1 hour and then returns home.

- Time: minutes
- Distance: kilometres

Draw the following graphs:

- Position–time
- Velocity–time
- Acceleration–time

Solution

Whenever we have a problem with several stages of motion, we must analyse each stage separately and determine the values of x, v, and a as functions of t.

Since this is uniform motion, we know that $a = 0\frac{m}{s^2}$ in every case.

Journey to School

The girl takes 20 minutes to reach the school.

- Distance travelled: 1 km
- Time taken:

$$20 \text{ min} = 20 \text{ min} \times \frac{1 \text{ h}}{60 \text{ min}} = \frac{20 \text{ h}}{60} = \frac{1 \text{ h}}{3} \approx 0.33 \text{ h}$$

- Velocity in km/min:

$$v = \frac{d}{t} = \frac{1 \text{ km}}{20 \text{ min}} = 0.05 \text{ km/min}$$

If we calculate the velocity in km/h:

$$v = \frac{d}{t} = \frac{1 \text{ km}}{0.33 \text{ h}} \approx 3 \text{ km/h}$$

Staying at School

The girl stays at school for 1 hour.

- Distance travelled: 0 km
- Time taken: 1 hour $= 1 \times 60$ min $= 60$ min
- Velocity:

$$v = \frac{d}{t} = \frac{0 \text{ km}}{1 \text{ h}} = 0 \text{ km/h}$$

Return Journey

The return journey also lasts 20 minutes.

- Distance travelled: 1 km
- Time taken: 20 min $= 0.33$ h
- Velocity: here we must be careful because she is returning home. The distance is still 1 km, but it is better to use displacement:

$$\Delta x = x_f - x_i = (0 - 1) \text{ km} = -1 \text{ km}$$

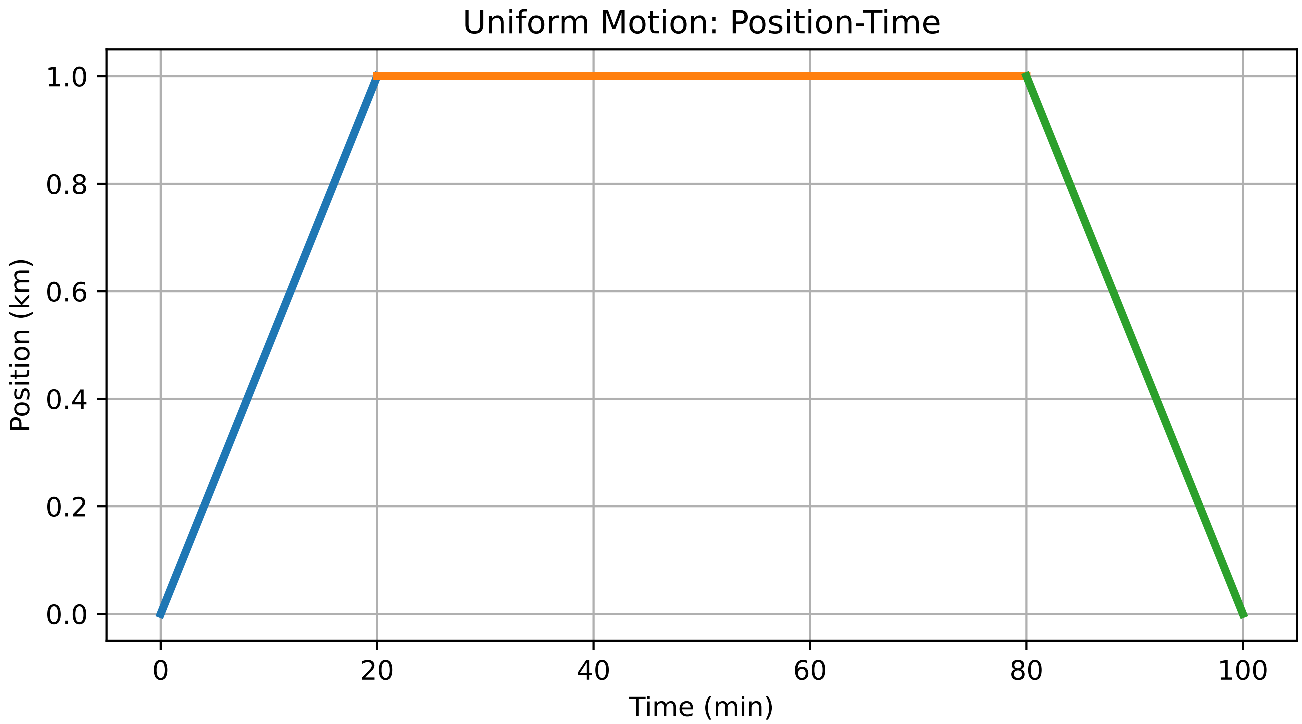
$$v = \frac{\Delta x}{t} = \frac{-1 \text{ km}}{20 \text{ min}} = -0.05 \text{ km/min} = -3 \text{ km/h}$$

Graphs and Their Interpretation

Interpreting a graph means understanding what we see and being able to explain it. Of course, it must match the information given in the problem.

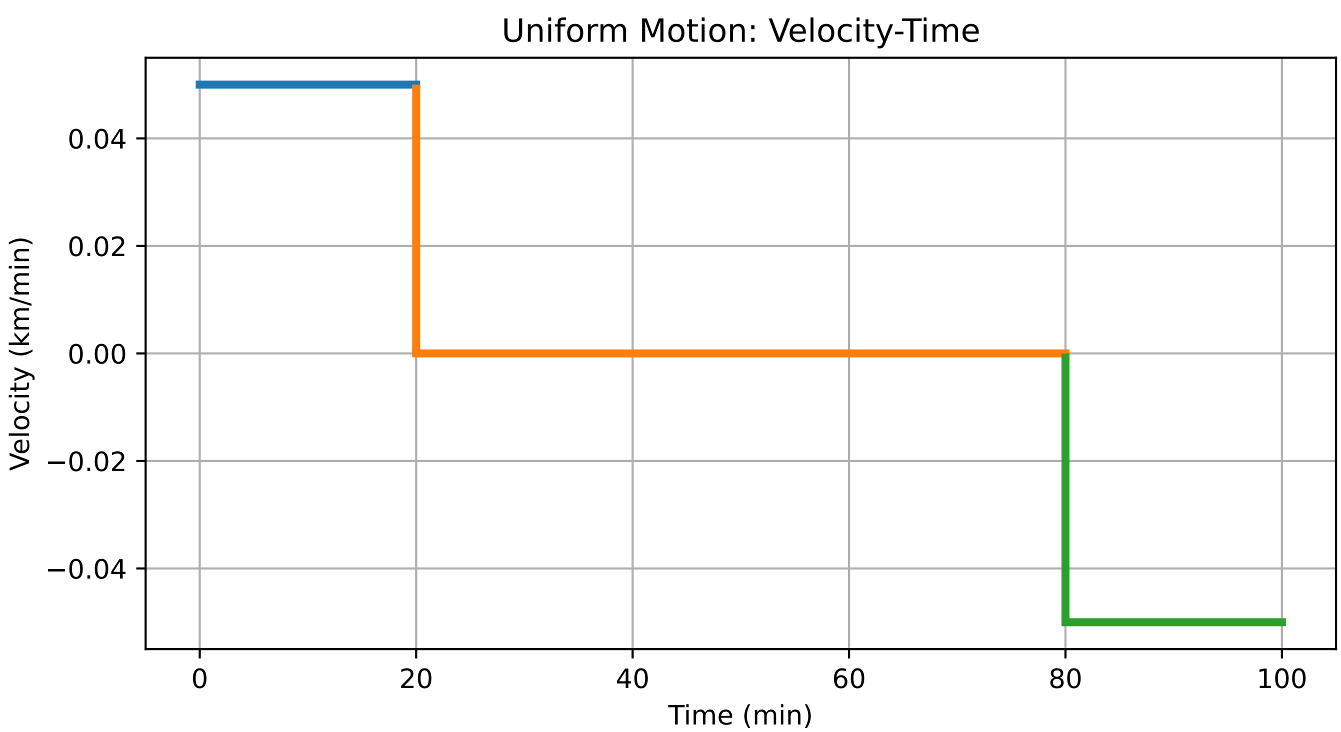
Position–Time Graph

- The position increases linearly from 0 km to 1 km during the journey to school.
- During the hour at school, the position remains constant.
- On the return journey, the position decreases linearly until she arrives back home.



Velocity–Time Graph

- The velocity is constant and positive during the journey to school.
- The velocity is 0 while she remains at school.
- The velocity is constant and negative during the return journey.



Acceleration–Time Graph

- The acceleration is 0 throughout the motion because the velocity is constant in each stage.